

2K Laser Powder bed Fusion Combustion Chamber and Specialty Alloy Build Processing
Statement of Work (SOW)
March 28, 2025

The National Aeronautics and Space Administration (NASA), specifically Marshall Space Flight Center (MSFC) has a need for laser powder bed fusion (L-PBF) component builds of novel additively manufactured (AM) NASA Nickel-based burn resistant alloys including OR-F1, ORCAAlloy-21, and Alloy 754-ODS, and additional oxide dispersion strengthened (ODS) based alloys such as GRX-810. Moreover, NASA has a need for GRCop-42 component builds and processing to support 2K-class nozzle testing scheduled to occur at the end of FY25.

Task 1, GRCop-42 2K-Class Combustion Chamber Builds Using L-PBF

1. The vendor shall complete a minimum of sixteen (16) chamber liner builds using GRCop-42.
2. NASA shall provide all models required for builds.
3. Vendor shall notify NASA of any errors or issues during builds.
4. Vendor shall notify NASA if powder is switched at any time to a different alloy during the NASA alloy builds.
5. Vendor shall complete powder removal after removal from the machine.

Task 2, Two (2) Month L-PBF machine lease to process GRX-810 and ORCAAlloy-21 for 2K-class injectors

1. The vendor shall provide a machine with parameters for builds using the burn resistant Ni-based alloys including OR-F1-M9 Alloy, ORCAAlloy-21, ORCAAlloy-14, Alloy 754-ODS, other ODS-based custom NASA alloys such as GRX-810, and standard AM alloys for comparison.
2. The machine shall be made available for a minimum 432 hours per month. An estimated minimum run time of 60% is estimated to maintain machine capacity.
3. The total period of development shall be 2 months of machine time shall be made available and include a kickoff meeting with technical points of contact (POC).
4. Vendor shall use custom developed parameters for the OR-F1 alloy developed under prior development work with NASA MSFC and Glenn Research Center (GRC).
5. Vendor shall notify NASA of any errors or issues during development builds.
6. Vendor shall notify NASA if powder is switched at any time to a different alloy during the NASA alloy development builds.
7. Vendor shall complete powder removal after removal from the machine.
8. Powder lots shall be tracked on a best effort basis to determine multiple uses (for additional studies on recycled powder).
9. Vendor shall provide build records and any powder certifications associated for traceability of builds.
10. Any excess powder shall be returned to NASA.
11. NASA shall provide ODS alloying of powder as needed. Vendor shall coordinate with NASA POC's to process powder.
12. NASA shall provide all sample geometry and models required for builds.

Task 3, Mechanical testing of samples to validate 2K material properties

1. The vendor shall complete a minimum of fifty (50) mechanical tests on samples from the development builds in Task 1.
2. The testing shall include mechanical tensile and low/high cycle fatigue. Test conditions should be coordinated with the NASA technical POC and agreed upon conditions and requirements.

Task 4, Machining of 2K Components for Assembly

1. The vendor shall coordinate with NASA to complete any machining per NASA's requirements for each part.

Task 5, Post-Processing of Chamber Channels Including Surface Enhancements

1. The vendor shall complete all heat treatment, stress relief, or other post processing techniques in accordance with NASA's requirements for each part.
2. The vendor shall coordinate with NASA to complete all require surface enhancements of the coolant channels in accordance with NASA's requirements.

Period of Performance (POP)

April 1, 2025 through December 10, 2025

Shipping

1. Contractor shall ship all deliverables and process development samples to:
Central Receiving, Bldg
4631 Marshall Space Flight
Center Huntsville, AL 35812

Deliverables

1. Parts built under Task 1.
2. Parts built under Task 2.
3. Test Reports from samples in Task 3.
4. Machined parts completed under Task 4.
5. Processed chambers completed under Task 5.